

# **Chapter Two**

## **Theory of Demand and Supply**

# Introduction

- how do markets work?
  - Why consumers demand certain things and not others?
  - why producers supply the things needed?
- how prices are determined in markets, and how the market can adjust naturally when there is disequilibrium?.
- what happens when there is change in a market and how change affects price and the quantity traded?
- The concept of elasticity is applied to show how the extent of change in markets can be measured.

# Chapter objectives

- **After covering this chapter, you will be able to:**
  - ✓ understand the concept of demand and the factors affecting it;
  - ✓ explain the supply side of a market and the determinants of supply;
  - ✓ analyze the causes of a shift in the demand curve (D) and the supply curve (S)
  - ✓ distinguish between the shift in the demand or supply curve and the movement along these curves.
  - ✓ understand how the market reaches equilibrium condition, and the possible factors that could cause a change in equilibrium and
  - ✓ explain the elasticity of demand and supply

# Theory of Demand

- **Demand** - quantity of a good or service that consumers are *willing and able* to buy at a *given market price* in a given time period, *ceteris paribus*.
- **Theory of demand** - an economic principle describing the relationship between *consumer demand for goods and services* and their *prices* in the market.

**Law of demand** - *price* of a commodity and its *quantity demanded* are *inversely related*.

- The higher the price of a product is, all else equal, the less of it will be demanded and vice versa

# The 'Ceteris paribus' Assumption

- **Ceteris paribus** is a Latin term widely used by economists to refer to a situation where 'other things remain equal' or are unchanged.
- Ceteris paribus allows economists to simplify a situation by assuming that apart from a single change of circumstances, everything else is unchanged. In this way, economists can model the effects of one change at a time.
- Remember to include the term in any written answers where relevant.

# Demand schedule

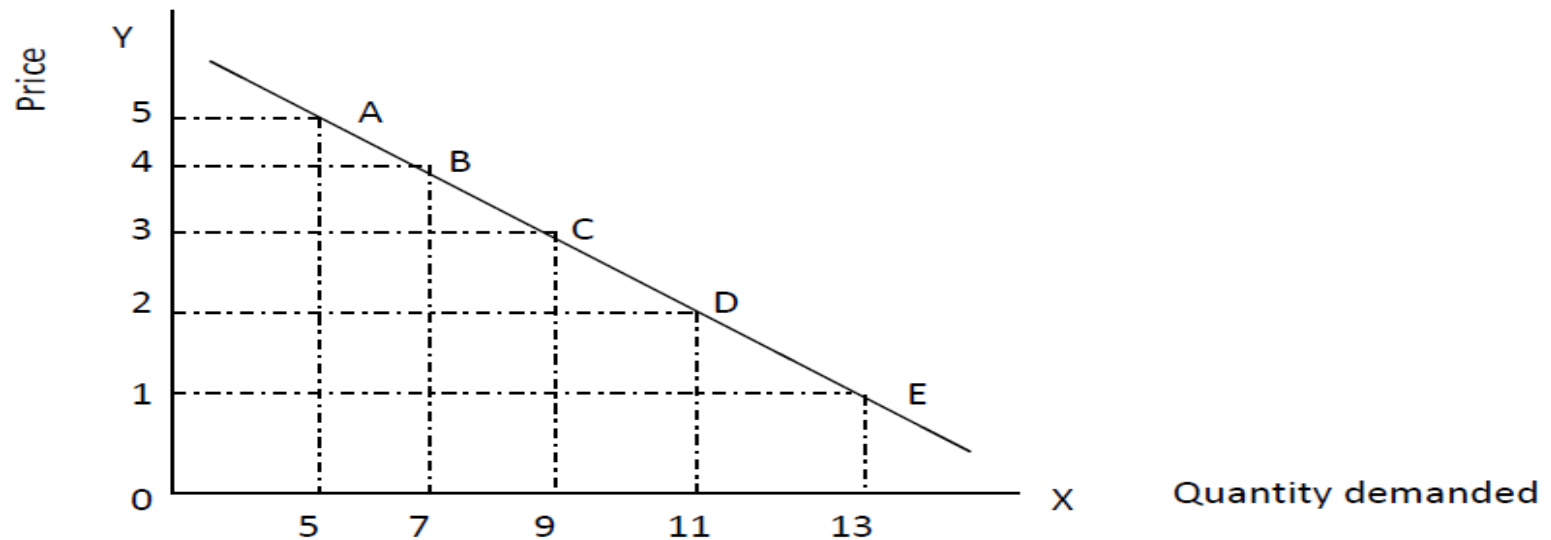
- **Demand schedule:** list of the various quantities of a commodity an individual consumer purchases at various levels of prices in the market.

| Combinations         | A | B | C | D  | E  |
|----------------------|---|---|---|----|----|
| Price per kg         | 5 | 4 | 3 | 2  | 1  |
| Quantity demand/week | 5 | 7 | 9 | 11 | 13 |

# Demand Curve

- **Demand curve** is a graphical representation of the relationship between different quantities of a commodity demanded by an individual at different prices per time period.

**Individual demand curve**



# Demand Function

**Demand function:** a mathematical relationship between price and quantity demanded, all other things remaining the same.

$$Q_d = f(P)$$

where  $Q_d$  is quantity demanded and  $P$  is price of the commodity

Example: Consider a linear demand function:  $Q = a + bp$

To determine the demand function for orange

- Calculate the slope of the demand curve:  $b = \frac{\Delta Q}{\Delta P} = \frac{7-5}{4-5} = -2$
- Determine the intercept (a) from  $7 = a - 2(4) \rightarrow a = 15$
- Thus,  **$Q = 15 - 2p$**



# Market Demand

- **Market Demand for a commodity** : derived by horizontally adding the quantity demanded for the product by all buyers at each price.

| Price | Individual demand |            |            | Market demand |
|-------|-------------------|------------|------------|---------------|
|       | Consumer-1        | Consumer-2 | Consumer-3 |               |
| 8     | 0                 | 0          | 0          | 0             |
| 5     | 3                 | 5          | 1          | 9             |
| 3     | 5                 | 7          | 2          | 14            |
| 0     | 7                 | 9          | 4          | 20            |

# Market Demand Curve

- Individual and market demand curve at price equal to 3.



# Market Demand Function

- **Example 1:** Suppose the individual demand function of a product is given by:  $Q = 20 - 2P$  and there are about 100 identical buyers in the market.

The market demand function is:

$$Q_m = 100 (20 - 2P) = 2000 - 200P$$

- **Example 2:** Individual demand function is give by  $P = 40 - Q/5$  and there 50 identical buyers in the market. Derive the market demand function ( $Q_m$ )

# Note

Do not confuse '***quantity demanded***' and '***demand***'.

Demand refers to the whole demand schedule and does not change when the price of the product changes. A price change causes a change in the quantity demanded.

- A change in the price of a good or service causes a change in the quantity demanded (movement *along* the demand curve).
- A change in a demand shifter causes a change in demand, which is shown as a *shift* of the demand curve.

Demand shifters include preferences, the prices of related goods and services, income, demographic characteristics, and buyer expectations.

# Determinants of demand

$$Q_{x,t} = f\{P_{x,t}, Y_t, P_{r,t}, P_{x,t+i}, Y_{t+i}, N, T\}$$

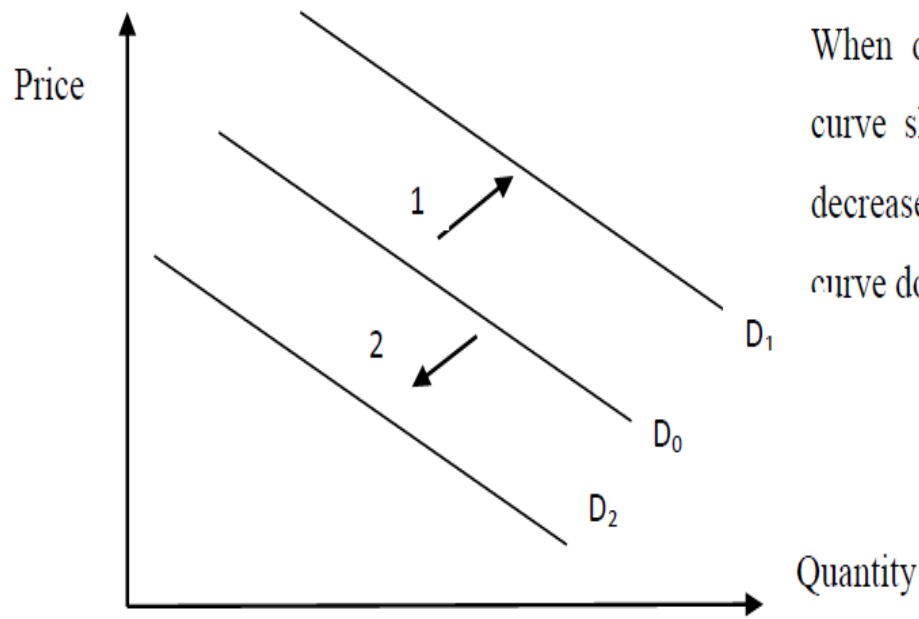
Where  $Q_{x,t}$  - Quantity demand of good x at period t,

- $P_{x,t}$  - Price of good x at period t
- $Y_t$  - Income of the consumers
- $P_{r,t}$  - Price of related goods (substitutes or complements)
- $P_{x,t+i}$  - Expected price of good x in some future period
- $Y_{t+i}$  - Consumer's expectation of income in future period
- $N$  - Number of buyers in the market
- $T$  - Taste or preference of consumers

# Change in Demand

- A change in any determinant of demand —**except for the good's price** results a **shift** in the **demand curve**.

## Shift in the demand curve



When demand increases, demand curve shifts upward ( $D_1$ ) while a decrease in demand shifts demand curve downwards ( $D_2$ ).

# Change in Demand

## □ Price of related goods

Two goods are said to be related if a change in the price of one good affects the demand for another good.

- ***Substitute goods*** are goods which satisfy the same desire of the consumer. If two goods are substitutes, **then price of one and the demand for the other are directly related.**
- ***Complimentary goods*** those goods which are jointly consumed. If two goods are complements, **then price of one and the demand for the other are inversely related.** Typical examples are cars and petrol

# Change in Demand

## Complements (coffee and doughnuts)



*Reducing the price of one...*



*increases the demand for the other.*



*Increasing the price of one...*



*reduces the demand for the other.*

## Substitutes (coffee and tea)



*Increasing the price of one...*



*increases the demand for the other.*



*Reducing the price of one...*



*reduces the demand for the other.*



# Change in Demand

## □ Income of the consumer

The effect of change in the income of a consumer on quantity demand of a good depends on the type of good.

- ***Normal goods*** are goods whose demand increases as income increase.
- ***Inferior goods*** are goods whose demand is inversely related with income. Eg. Low quality goods (noodles), second hand clothes,...

# Change in Demand

| Normal Good                                                                         |                                 |                                                                                       |                          |
|-------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------|--------------------------|
|    | An <i>increase</i> in income... |    | <i>increases</i> demand. |
|    | A <i>decrease</i> in income...  |    | <i>decreases</i> demand. |
| Inferior Good                                                                       |                                 |                                                                                       |                          |
|   | An <i>increase</i> in income... |   | <i>decreases</i> demand. |
|  | A <i>decrease</i> in income...  |  | <i>increases</i> demand. |

# Change in Demand

## □ Consumer expectation of income and price

- **Price Expectation:** Higher price expectation will increase current while a lower future price expectation will decrease the demand for the good.
- **Income Expectation:** Higher expectation of future income increase current demand while lower expectation of future income decrease demand.

# Change in Demand

## ❑ Number of buyer in the market

- Since market demand is the horizontal sum of individual demand, an increase in the number of buyers will increase demand while a decrease in the number of buyers will decrease demand.

## ❑ Taste or preference of the consumer

- When the taste of a consumer changes in favor of a good, her/his demand will increase and vice versa.

# Possible causes for shifts in the demand curve.

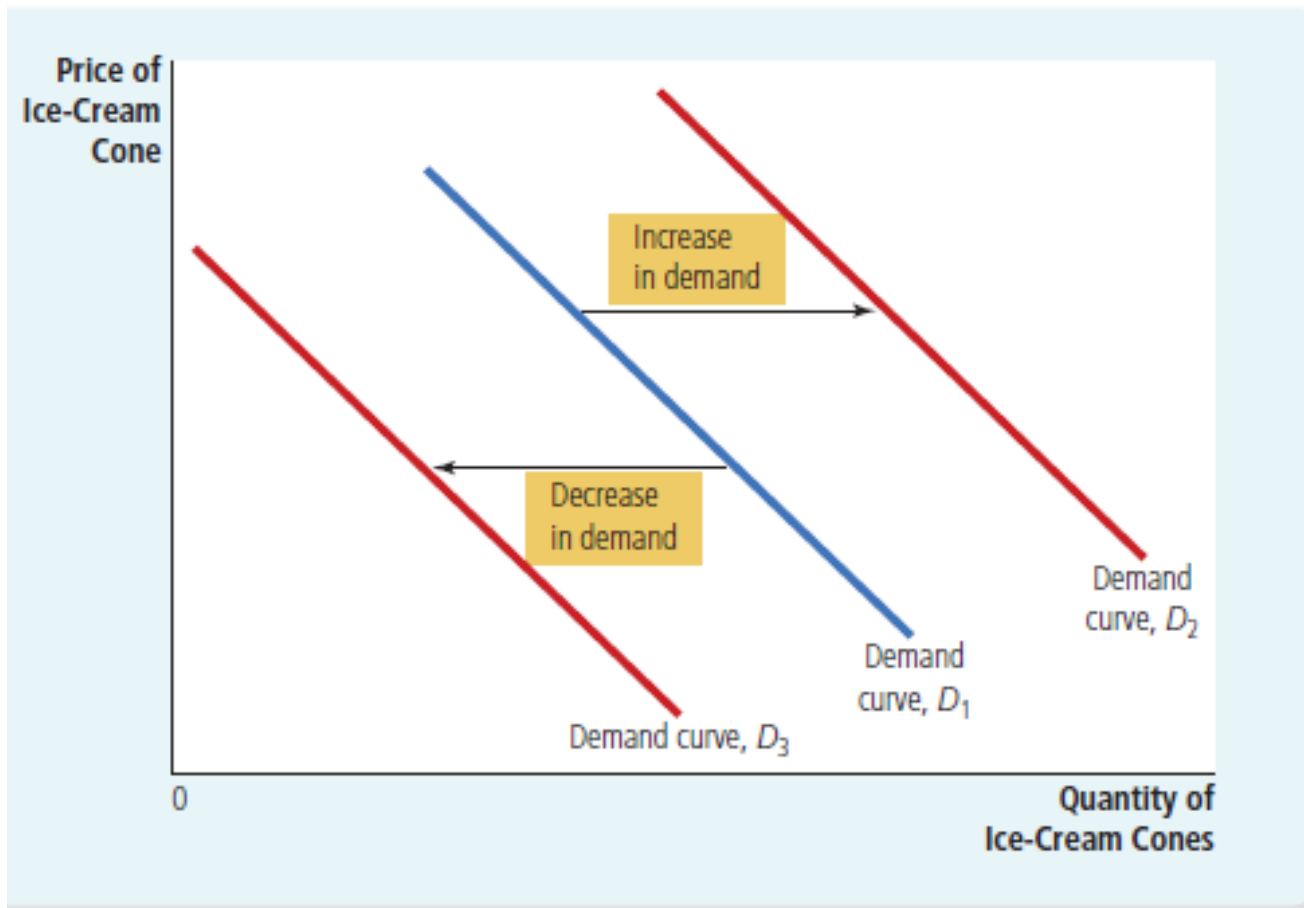
## A shift to the right in the demand curve

- an increase in income
- an increase in the price of substitutes
- a decrease in the price of complements
- a favourable change in fashion, taste and attitudes.

## A shift to the left in the demand curve

- a decrease in income
- a decrease in the price of substitutes
- an increase in the price of complements
- an unfavourable change in fashion, taste and attitudes.

## Example: A shift in the Demand Curve



# Elasticity of demand

- **Elasticity of demand** refers to the **degree of responsiveness of quantity demanded** of a good to a change in **its price**, or change in **income**, or change in **prices of related goods**.
- There are three kinds of demand elasticity:
  - Price elasticity
  - Income elasticity
  - Cross-price elasticity

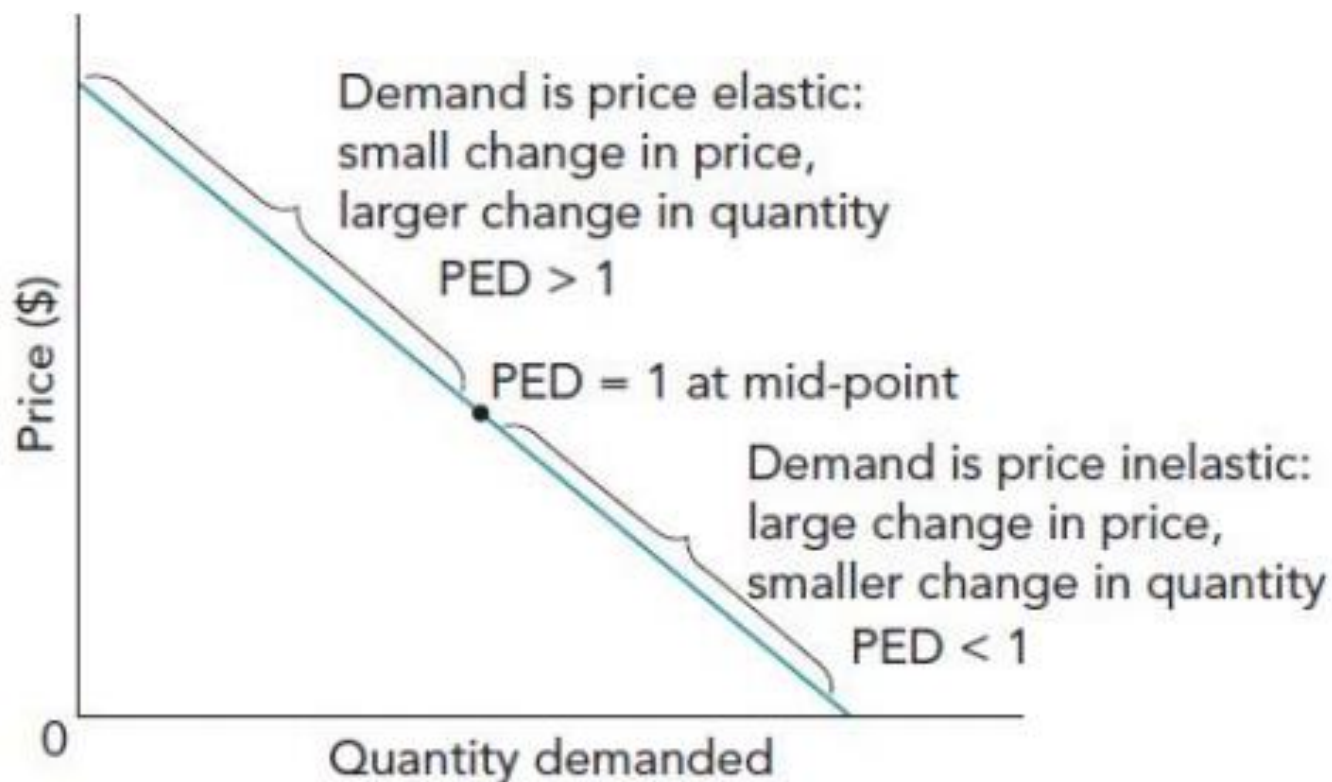
# Price Elasticity of Demand

- Measures degree of responsiveness of demand to change in price.
- It is computed as the percentage change in quantity demanded divided by the percentage change in price.
  - i) If  $|\varepsilon| > 1$ , demand is said to be elastic and the product is luxury product
  - ii) If  $0 \leq |\varepsilon| < 1$ , demand is inelastic and the product is necessity
  - iii) If  $|\varepsilon| = 1$ , demand is unitary elastic.
  - iv) If  $|\varepsilon| = 0$ , demand is said to be perfectly inelastic.
  - v) If  $|\varepsilon| = \infty$ , demand is said to be perfectly elastic.



# Price Elasticity of Demand (PED) and a downward-sloping linear demand curve

- PED changes along a demand curve



# Point and arc elasticity of demand

❑ **Point elasticity of demand:** is calculated to find elasticity at a given point.

$$E_d^P = \frac{\text{Percentage change in quantity demanded}}{\text{percentage change in price}} = \frac{\% \Delta Q_d}{\% \Delta P}$$

$$\text{where } \% \Delta Q_d = \frac{Q_1 - Q_0}{Q_0} \times 100 \text{ and}$$

$$\% \Delta P = \frac{P_1 - P_0}{P_0} \times 100$$

$$\text{Thus, } E_d^P = \frac{\frac{Q_1 - Q_0}{Q_0} \times 100}{\frac{P_1 - P_0}{P_0} \times 100} = \frac{Q_1 - Q_0}{P_1 - P_0} \cdot \frac{P_0}{Q_0} = \frac{\Delta Q}{\Delta P} \cdot \frac{P_0}{Q_0}$$

# Point and arc elasticity of demand

- ❑ **Arc elasticity of demand:** calculates elasticity by taking midpoints of the old and the new values of both price and quantity demanded.

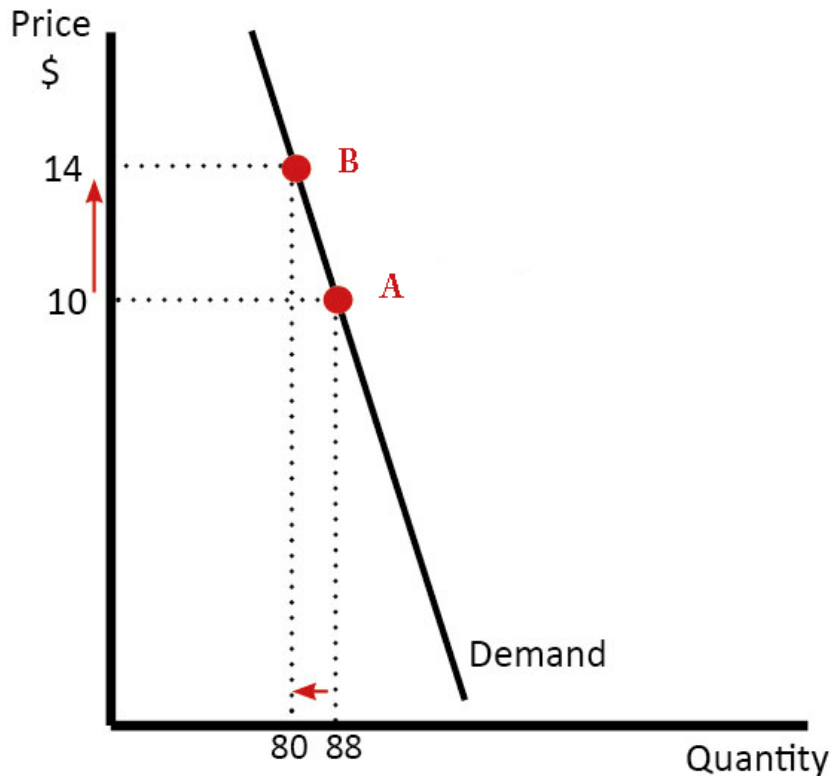
$$E_d = \frac{\text{Change in quantity demanded}}{\text{Original quantity plus new quantity demanded}} \div \frac{\text{Change in price}}{\text{Original price plus new price}}$$

Symbolically, the formula may be expressed thus:

$$E_d = \frac{Q_1 - Q_0}{Q_0 + Q_1} \div \frac{P_1 - P_0}{P_0 + P_1}$$

# Point and arc elasticity of demand

## • Example 1



[www.economicshelp.org](http://www.economicshelp.org)

### Point elasticity from point A to B.

- $\% \Delta Q = 8/88 = -10\%$
- $\% \Delta P = 4/10 = 40\%$

$$PED = -10/40 = \mathbf{-0.25}$$

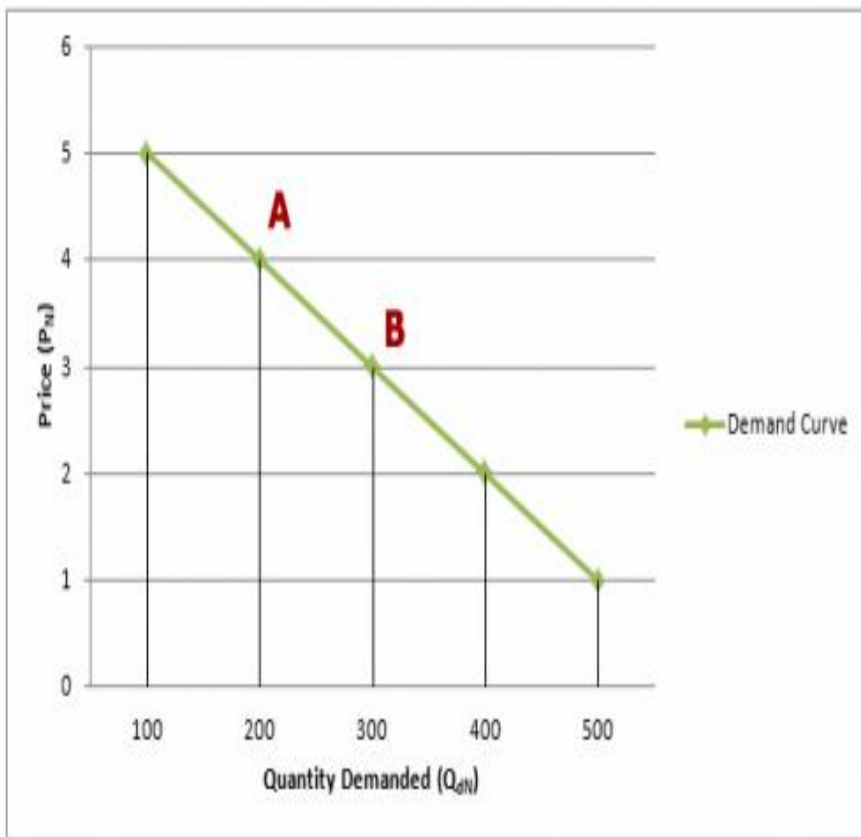
### Arc elasticity of demand

- The mid point of Q =  $(80+88)/2 = 84$
- The mid-point of P =  $(10+14)/2 = 12$
- $\% \Delta Q = 80-88/84 = -0.09524$
- $\% \Delta P = (14-10)/12 = 0.3333$

$$PED = 0.333/-0.9524 = \mathbf{-0.285}$$

# Point and arc elasticity of demand

- **Example 2**



- **Point elasticity A to B**

- Quantity increase from 200 to 300 =  $100/200 = 50\%$
- Price falls from 4 to 3 =  $1/4 = -25\%$
- $PED = 50 / -25 = -2$

- **Mid Point (Arc) Elasticity A to B**

- Mid point of Q =  $(200+300)/2 = 250$
- Mid Point of P =  $(3+4) / 2 = 3.5$
- $\% Q = (100/250) = 40\%$
- $\% P = 1/3.5 = 28.57$

$$PED = 40 / -28.57 = -1.4$$

$$\text{or } (3.5/250) * 100/1 = -1.4$$

# Relating Elasticity to Changes in Total Revenue

| Price Elastic                                                                       |                                |                                                                                                                    |
|-------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------|
|    | An <i>increase</i> in price... |  <i>reduces</i> total revenue.   |
|    | A <i>reduction</i> in price... |  <i>Increases</i> total revenue. |
| Total revenue moves in the direction of the quantity change.                        |                                |                                                                                                                    |
| Price Inelastic                                                                     |                                |                                                                                                                    |
|    | An <i>increase</i> in price... |  <i>Increases</i> total revenue. |
|    | A <i>reduction</i> in price... |  <i>reduces</i> total revenue.   |
| Total revenue moves in the direction of the price change.                           |                                |                                                                                                                    |
| Unit price Elastic                                                                  |                                |                                                                                                                    |
|   | An <i>increase</i> in price... |  no change in total revenue.   |
|  | A <i>reduction</i> in price... |  no change in total revenue.   |
| Total revenue does not change as price changes.                                     |                                |                                                                                                                    |

# Determinants of Price Elasticity of Demand

- **The availability of substitutes:** the more substitutes available for a product, the more *elastic* will be the price elasticity of demand.
- **Time:** in the long- run, price elasticity of demand tends to be elastic.
  - More substitute goods could be produced.
  - People tend to adjust their consumption pattern.
- **The proportion of income consumers spend for a product:** the smaller the proportion of income spent for a good, the less price elastic will be.
- **The importance of the commodity in the consumers' budget :**
  - Luxury goods: tend to be more **elastic**, example: gold.
  - Necessity goods: tend to be **less elastic** example: Salt.

# Income Elasticity of Demand

- Measure of responsiveness of demand to change in income.

$$\varepsilon_d^I = \frac{\% \Delta Q_d}{\% \Delta I} = \frac{\Delta Q}{\Delta I} \cdot \frac{I}{Q}$$

Point income elasticity of demand:

- i) If  $\varepsilon_d^I > 1$ , the good is **luxury** good.
- ii) If  $\varepsilon_d^I < 1$  (and positive), the good is **necessity** good,
- iii) If  $\varepsilon_d^I < 0$ , (negative), the good is **inferior** good.



# Income Elasticity of Demand

- **Example 1:** Suppose initially the income of the consumer were 1000 birr and s/he demands 50 units of good x. Now, assume that the income of the consumer increases to 1500 birr and his/her demand for good x becomes 70 units. Calculate the income elasticity of demand and determine what kind of good 'x' is?

$$\varepsilon_d^i = \frac{70-50}{1500-1000} \times \frac{1000}{50} = 0.8, \text{ X is necessity good}$$

- **Example 2:** Suppose the average real income of customers of good Z falls from 50,000 birr to 40,000 birr, and the demand for good Z decreases from 10,000 to 5,000 units sold, all other things unchanged.

# Cross price Elasticity of Demand

- The cross-price elasticity of demand measures how the quantity demanded of one good responds to a change in the price of another good.

$$\epsilon_{xy} = \frac{\% \Delta Q_x}{\% \Delta P_y} = \frac{Q_{x_1} - Q_{x_0}}{P_{y_1} - P_{y_0}} \cdot \frac{P_{y_0}}{Q_{x_0}}$$

- i) The cross – price elasticity of demand for substitute goods is **positive**.
- ii) The cross – price elasticity of demand for complementary goods is **negative**.
- iii) The cross – price elasticity of demand for unrelated goods is **zero**.

# Cross price Elasticity of Demand

- **Example 1:** Consider the following data and calculate the cross –price elasticity of demand between the two goods. What can you say about the two goods?

| Unit price of Y | Quantity demanded of X |
|-----------------|------------------------|
| 10              | 1500                   |
| 15              | 1000                   |

$$\varepsilon_{xy} = \frac{\Delta Q_x}{\Delta P_y} * \frac{P_{y_o}}{Q_{x_o}} = \left( \frac{1000 - 1500}{15 - 10} \right) * \frac{10}{1500} = \frac{-500}{5} * \frac{10}{1500} = -0.67$$

- **X and Y are complements**

- **Example 2:** Consider the following table and calculate the cross price elasticity of demand for y with respect to the price of x.

| Price of X | Qty dd of X | Qty dd of Y |
|------------|-------------|-------------|
| 30         | 400         | 250         |
| 20         | 700         | 150         |

# **THEORY OF SUPPLY**

# Theory of Supply

- **Supply**: various quantities of a product that sellers or producers are **willing and able** to provide at a given **market price** in a particular period of time.
- **Theory of Supply**: an economic principle describing the relationship between **producers supply of goods and services** and their **prices** in the market.
- **The law of supply**: **Price** and **quantity** supplied has a **positive** relationship.
  - It states that, ceteris paribus, as **price** of a product **increase**, **quantity supplied** of the product **increases**, and **vice versa**.

# Supply schedule

**Supply schedule:** is a list of the various quantities of a commodity an individual producer (firm) offered for sale at different levels of market prices.

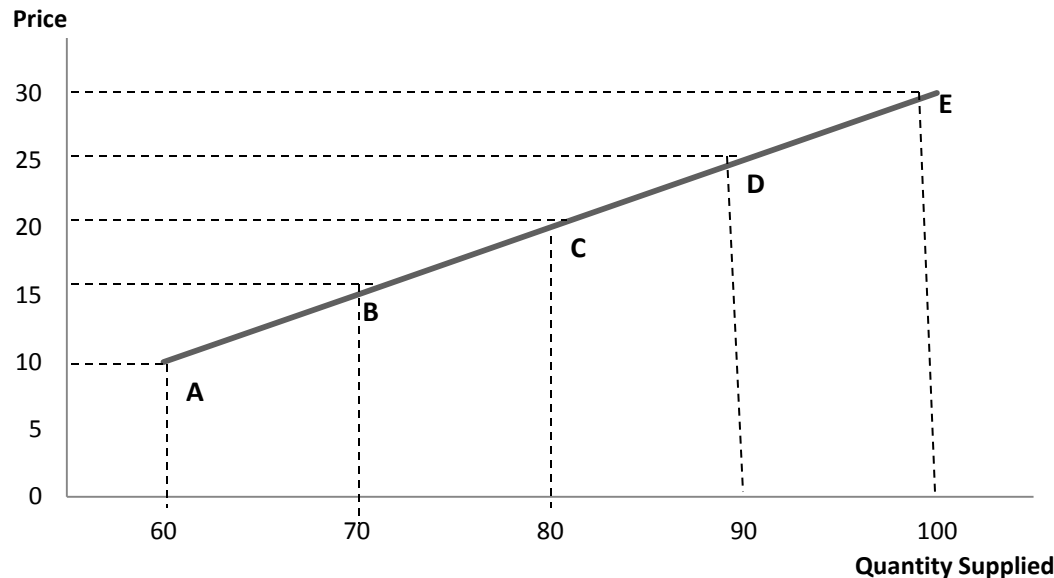
- **Example:** An individual seller's supply schedule for orange

|                           |     |    |    |    |    |
|---------------------------|-----|----|----|----|----|
| Price ( birr per kg)      | 30  | 25 | 20 | 15 | 10 |
| Quantity supplied kg/week | 100 | 90 | 80 | 70 | 60 |

# Supply Curve

**Supply curve** is a graphical representation of the relationship between different quantities of a commodity supplied by an individual producer at different prices per time period.

- **Supply Curve of a producer**



# Supply Function

**Supply function:** is a mathematical relationship between price and quantity supplied, all other things remaining the same.

$$Q_s = f(P)$$

where  $Q_s$  is quantity supplied and  $P$  is price of the commodity

Example: Consider a linear supply function:  $Q_s = c + dp$

To determine the supply function for orange

- Calculate the slope of the supply curve (d)
- Determine the intercept (c)

$$b = \frac{\Delta Q}{\Delta P} = \frac{70-60}{15-10} = 2$$

$$70 = a + 2(15) \rightarrow a = 40$$

$$Q_s = \mathbf{40 + 2p}$$



# Market Supply

**Market supply:** It is derived by horizontally adding the quantity supplied of the product by all sellers at each price.

- Sellers Supply of good x**

| Price per unit | Quantity supplied by seller 1 | Quantity supplied by seller 2 | Quantity supplied by seller 3 | Market supply per week |
|----------------|-------------------------------|-------------------------------|-------------------------------|------------------------|
| 5              | 11                            | 15                            | 8                             | 34                     |
| 4              | 10.5                          | 13                            | 7                             | 30.5                   |
| 3              | 8                             | 11.5                          | 5.5                           | 25                     |
| 2              | 6                             | 8.5                           | 4                             | 18.5                   |
| 1              | 4                             | 6                             | 2                             | 12                     |

# Determinants of supply

$$Q_{x,t}^S = f\{P_{x,t}, P_{in,t}, P_{r,t}, P_{x,t+i}, T\&S, N, W, Tec \dots\}$$

where

- $Q_{x,t}^S$  - Quantity supply of good x at period t,
- $P_{x,t}$  - Price of good x at period t
- $P_{in,t}$  - Price of inputs at time t( cost of inputs)
- $P_{r,t}$  - Prices of related goods at time t
- $P_{x,t+i}$  - Sellers' expectation of price of the product
- $T\&S$  - Taxes & subsidies
- $N$  - Number of sellers in the market
- $W$  - Weather
- $Tec$  - Technology

# Determinants of supply

## ❑ Change in input price

- An increase in the price of inputs such as labor, raw materials, capital, etc. causes a decrease in the supply of the product which is represented by a leftward shift of the supply curve. Likewise, a decrease in input price causes an increase in supply.

## ❑ Price of related goods

- If the price of a **substitute good in production** rises, then the firm increases the supply of the goods that have a higher price. This leads to a decrease in the supply of the goods that have a lower price.
- An increase in the price of a **complementary good in production** causes an increase in supply. With the higher price, sellers sell more of the complementary good, resulting in increased production and supply of this good as well.

# Determinants of supply

## ❑ Suppliers' Expectations

- Higher price expectation by suppliers in the future reduces the current supply of the good and vice versa.

## ❑ Taxes and Subsidies

- Taxes reduces profits, therefore increase in taxes reduce supply whereas decrease in taxes increase supply. Subsidies reduce the burden of production costs on suppliers, thus increasing the profits. Therefore increase in subsidies increase supply and decrease in subsidies decrease supply.

# Determinants of supply

## ❑ Change in weather condition

- A change in weather condition will have an impact on the supply of a number of products, esp. agricultural products. Eg., Good weather condition boosts the supply of agricultural products, other things remaining unchanged. This shifts the supply curve of a given agricultural product outward. Bad weather condition will have the opposite impact.

## ❑ Change in Technology

- Technological advancement enables a firm to produce and supply more in the market. This shifts the supply curve outward.

# Possible causes for shifts in the supply curve.

## A shift to the **right** in the supply curve

- a decrease in costs of production
- growth in the size of the industry
- a decrease in the price of competitor's goods
- decrease in an indirect tax or increase in subsidy.

## A shift to the **left** in the supply curve

- an increase in costs of production
- decline in the size of the industry
- an increase in the price of competitor's goods
- increase in an indirect tax or fall in subsidy.

# Elasticity of supply

- Measures the degree of responsiveness of supply to change in price

$$E_S = \frac{\% \text{ change in quantity supplied}}{\% \text{ change in price}}$$

- It can be calculated using both **point** and **arc method**.

$$E_S^P = \frac{\Delta Q}{Q} \div \frac{\Delta P}{P} = \frac{\Delta Q}{\Delta P} * \frac{P}{Q}$$

$$E_S^A = \frac{Q_1 - Q_0}{Q_0 + Q_1} \div \frac{P_1 - P_0}{P_0 + P_1} = \frac{\Delta Q}{\Delta P} * \frac{P_0 + P_1}{Q_0 + Q_1}$$

- Price elasticity of supply can be **elastic**, **inelastic**, **unitary elastic**, **perfectly elastic** or **perfectly inelastic**.

# Determinants of elasticity of supply

- **Length and complexity of production:** the less complex and the shorter to produce a good, the more elastic will be its supply, and vice versa
- **Mobility/availability of factors:** If the factors of production are easily available and if a producer producing one good can switch their resources and put it towards the creation of a product in demand, then it can be said that the PES is relatively elastic. The inverse applies to this, to make it relatively inelastic.
- **Time to respond:** The more time a producer has to respond to price changes the more elastic the supply. Supply is normally more elastic in the long run than in the short run for produced goods.
- **Inventories:** A producer who has a supply of goods or available storage capacity can quickly increase supply to market making supply more elastic.



# Determinants of elasticity of supply

- **Changes in marginal cost of production.** If with the expansion of output, marginal cost increases and marginal return declines, the price elasticity of supply will be less elastic to that extent.
- **Spare or excess production capacity:** A producer who has unused capacity can (and will) quickly respond to price changes in his market assuming that variable factors are readily available, resulting in more elastic supply.
- **Availability of infrastructure facilities:** If infrastructure facilities are available for expanding output of a particular good in response to the rise in prices, the elasticity of supply will be relatively more elastic.
- **Easiness of entering in the market :** goods operating in markets of free/easy entrance have relatively more elastic supply.

## 2.3. Market equilibrium

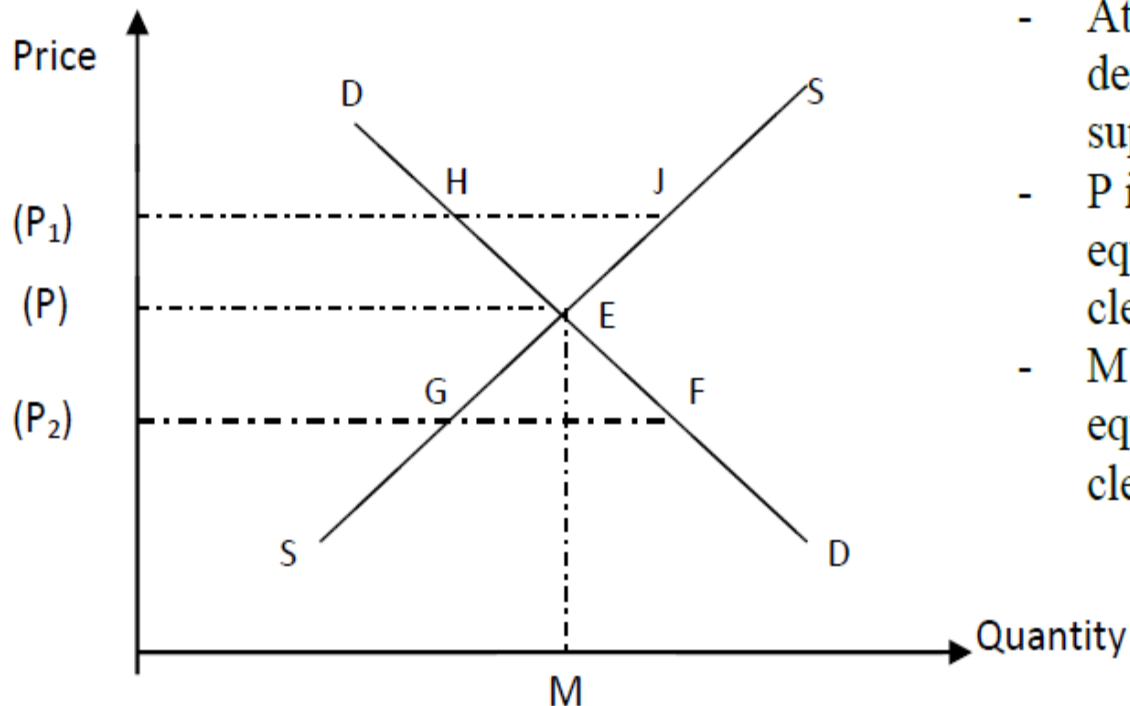
(Supply and Demand Together)

$$Q_d = Q_s$$

# 2.3 Market equilibrium

## (Supply and Demand Together)

Market equilibrium occurs when market demand equals market supply.

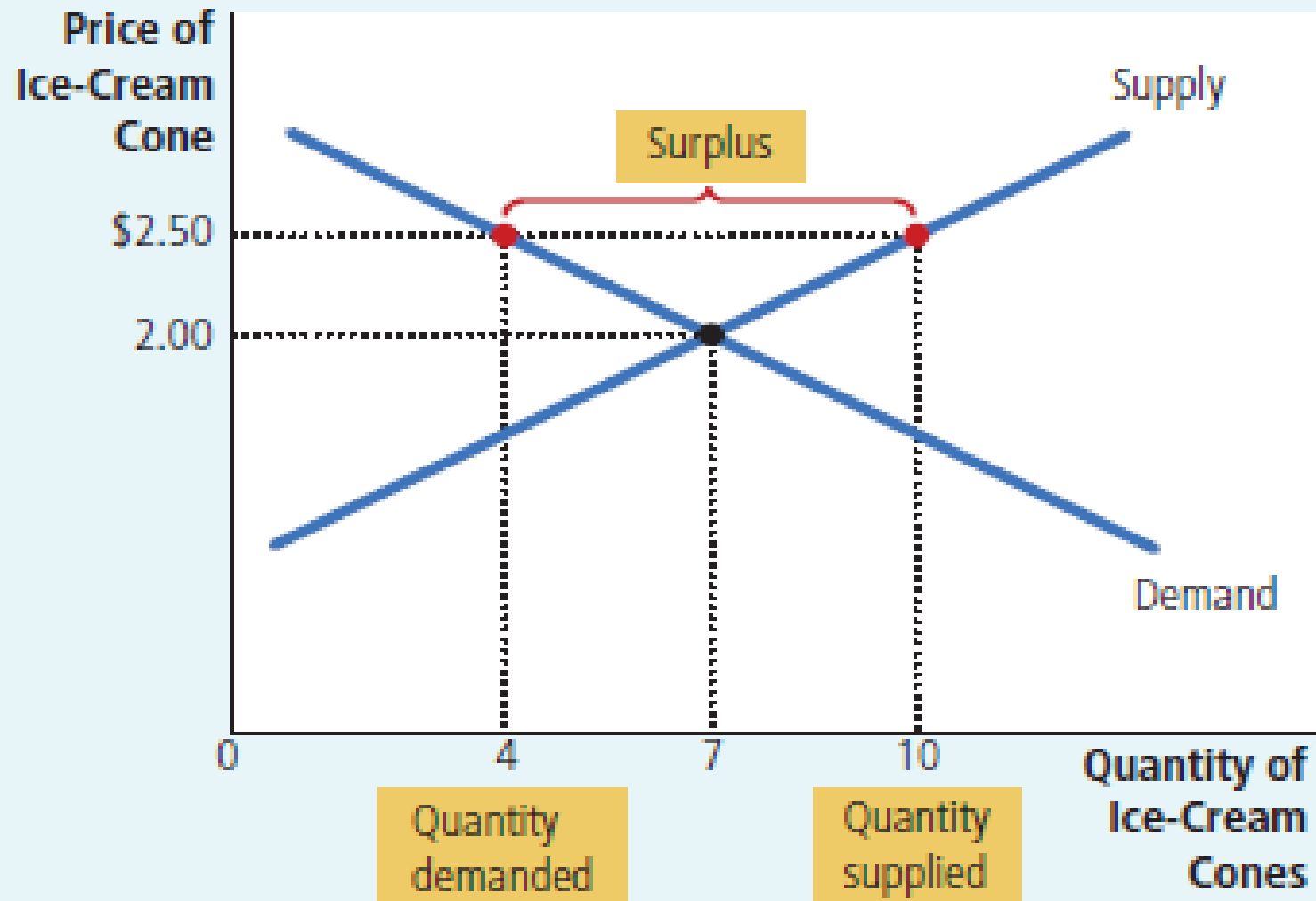


- At point 'E' market demand equals market supply (equilibrium point)
- P is the market equilibrium (market clearing) price.
- M is the market equilibrium (market clearing) quantity.

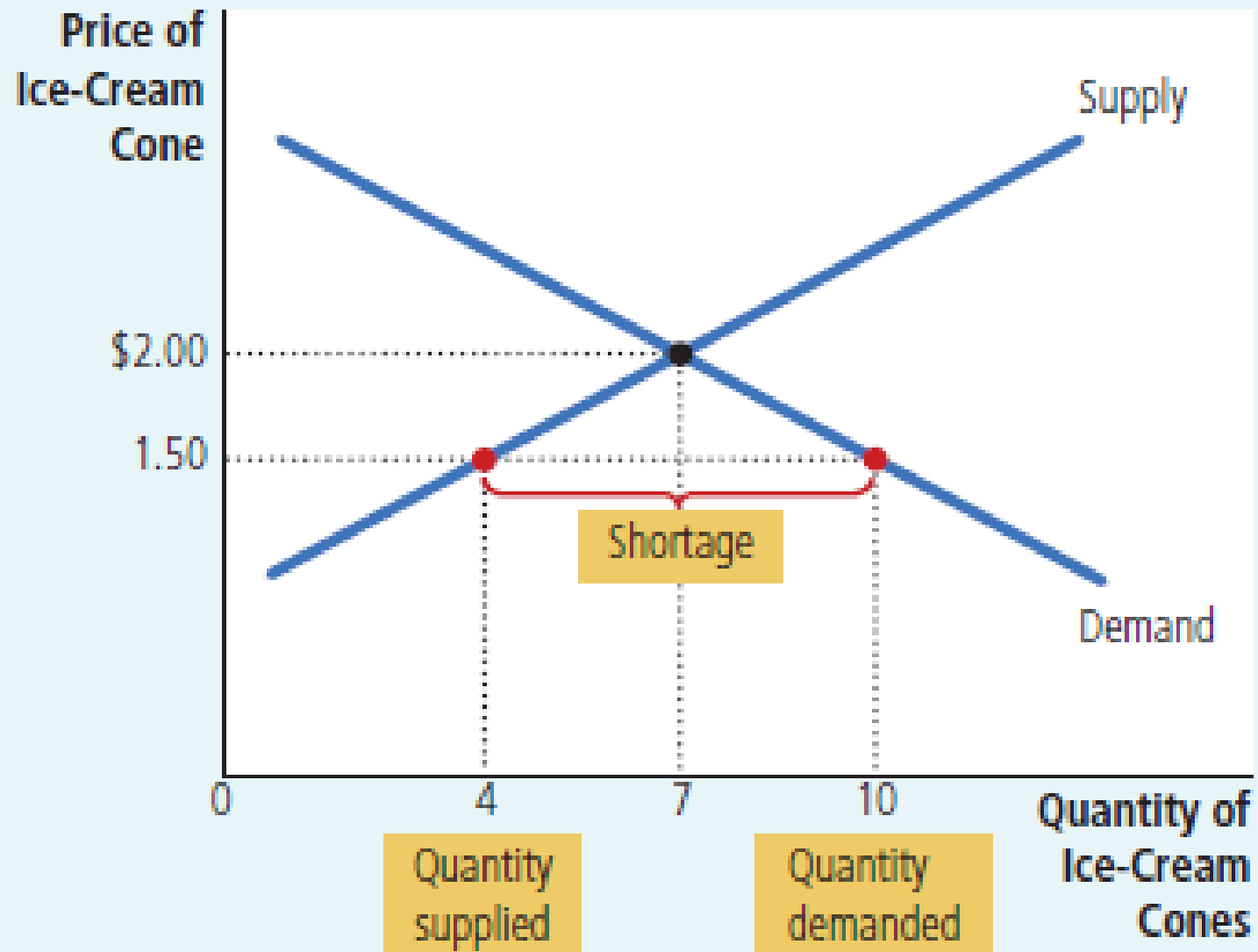
# Surplus and Shortage

- **Surplus:** A situation in which quantity supplied is greater than quantity demanded.
  - A surplus is sometimes called a situation of *excess supply*.
- **Shortage:** a situation in which quantity demanded is greater than quantity supplied.
  - A shortage is sometimes called a situation of *excess demand*.

(a) Excess Supply



### (b) Excess Demand



# Market equilibrium

**Example:** Given market demand:

$$Q_d = 100 - 2P, \text{ and market supply: } P = (Q_s / 2) + 10$$

- a) Calculate the market equilibrium price and quantity
- b) Determine, whether there is surplus or shortage at  $P = 25$  and  $P = 35$ .

a) At equilibrium,  $Q_d = Q_s$

$$100 - 2P = 2P - 20$$

$$4P = 120$$

$$\underline{\underline{\bar{P} = 30}}, \text{ and } \underline{\underline{\bar{Q} = 40}}$$

b)  $Q_d(\text{at } P = 25) = 100 - 2(25) = 50$  and  $Q_s(\text{at } P = 25) = 2(25) - 20 = 30$

Therefore, there is a shortage of:  $50 - 30 = \underline{\underline{20 \text{ units}}}$

$Q_d(\text{ at } P=35) = 100 - 2(35) = 30$  and  $Q_s(\text{ at } p = 35) = 2(35) - 20 = 50$ , a surplus of **20 units**

# Market equilibrium

## Exercise( Try it!)

1. The market demand for a good is characterized by the function:

$$Q_d = 1100 - 50P$$

and the market supply is defined by:

$$Q_s = 100P - 1000$$

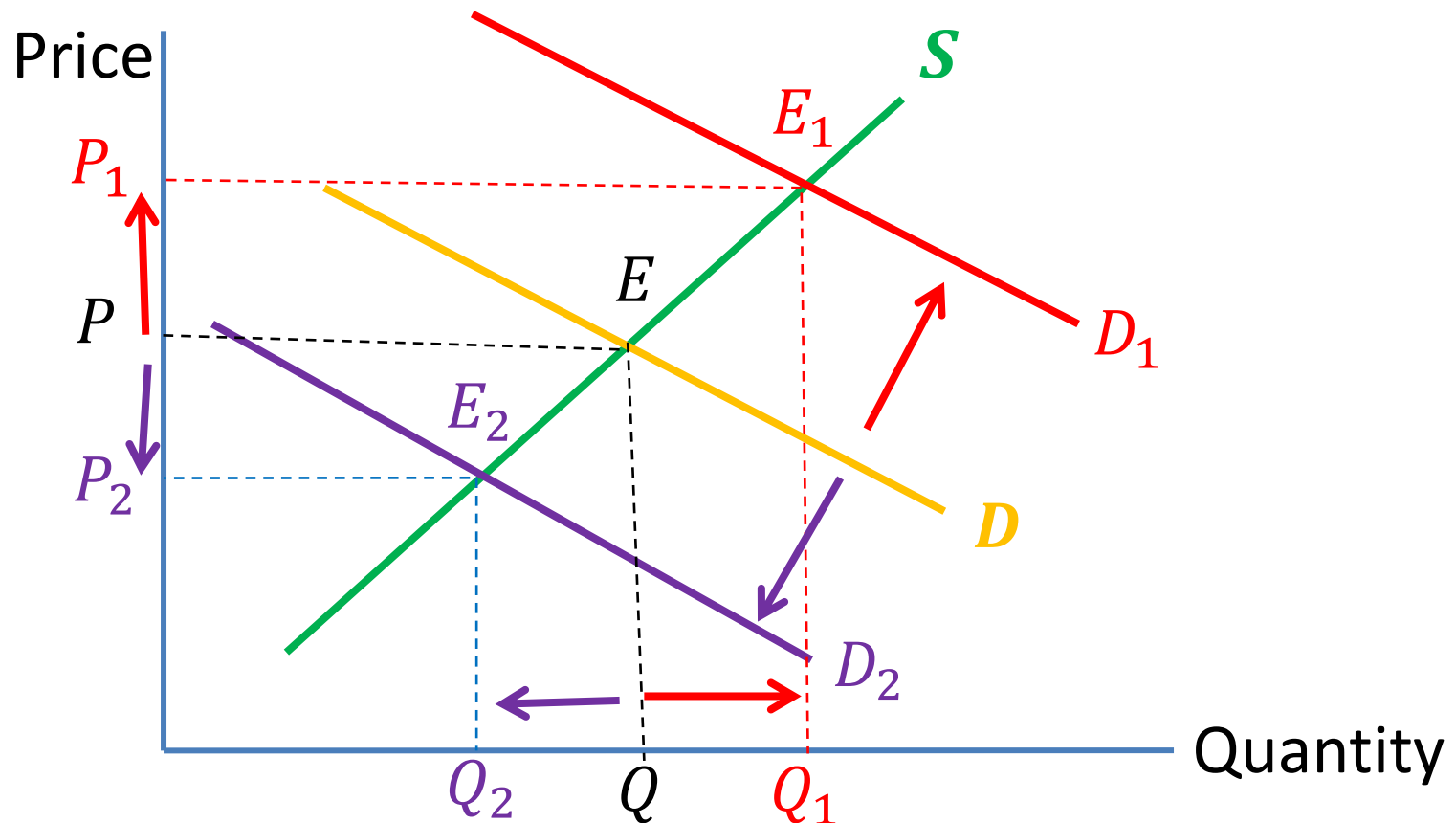
- a) Calculate the current **market equilibrium price** and **quantity**, and illustrate this equilibrium on **a graph**.
- b) Calculate the price elasticity of demand at the equilibrium point and interpret it.
- c) Calculate the price elasticity of supply at the equilibrium point and interpret it.
- d) What would be the state of the market if market price was fixed at Birr 16 per unit?



# Effects of Shift in Demand and Supply on Equilibrium Price and Quantity

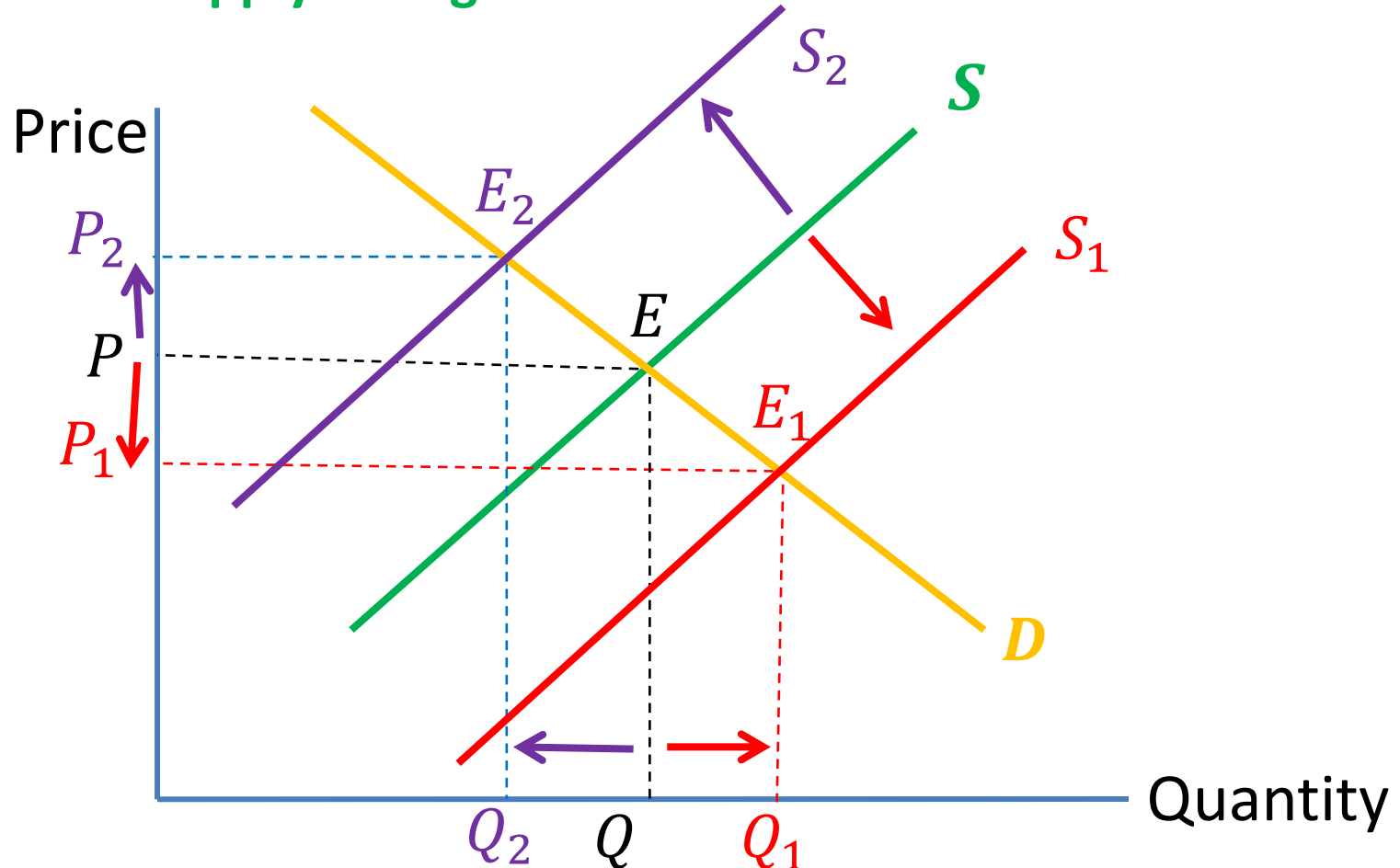
Changes in demand and supply bring about changes in the equilibrium price level and the equilibrium quantity.

## a) When Demand Changes and Supply Remains Constant



## Effects of Shift in Demand and Supply on Equilibrium Price and Quantity

b) When Supply Changes and Demand Remains constant



# Effects of Shift in Demand and Supply on Equilibrium Price and Quantity

## c) Effects of Combined Changes in Demand and Supply

- When **both demand** and **supply increase**, the **equilibrium quantity** of the product will **increase** definitely.
- But, it is **not certain** whether the **price** will **rise** or **fall**.
  - If an increase in demand is more than an increase in supply, then the price goes up.
  - If an increase in supply is more than an increase in demand, the price falls but the quantity increases.
  - If the increase in demand and supply is same, then the price remains the same.
- The same principle applies for a decreases in both demand and supply.
- For a summary, look at the following table.

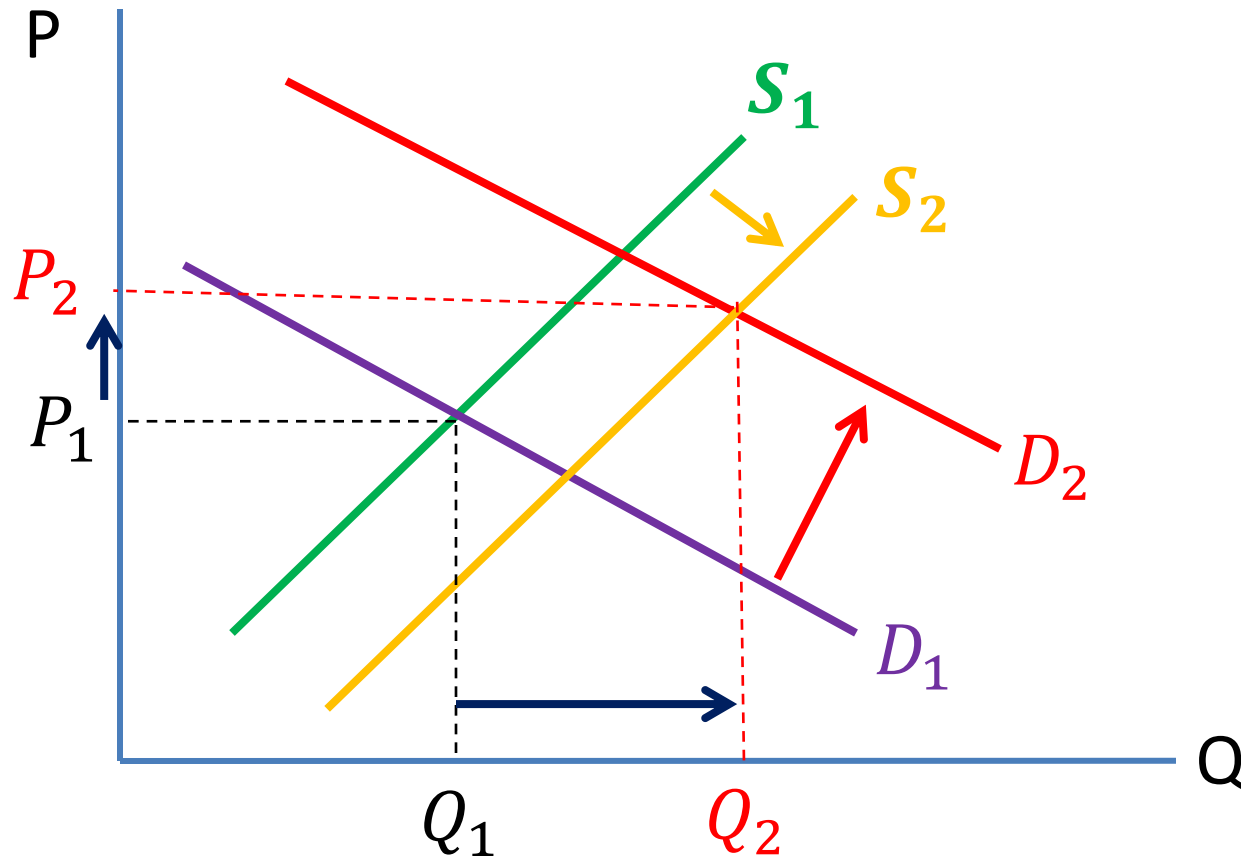
# What Happens to Price and Quantity When Supply or Demand or both Shifts?

|                          | No Change<br>in Supply | An Increase<br>in Supply | A Decrease<br>in Supply |
|--------------------------|------------------------|--------------------------|-------------------------|
| No Change<br>in Demand   | P same<br>Q same       | P down<br>Q up           | P up<br>Q down          |
| An Increase<br>in Demand | P up<br>Q up           | P ambiguous<br>Q up      | P up<br>Q ambiguous     |
| A Decrease<br>in Demand  | P down<br>Q down       | P down<br>Q ambiguous    | P ambiguous<br>Q down   |

# What Happens to Price and Quantity When Supply or Demand or both Shifts?

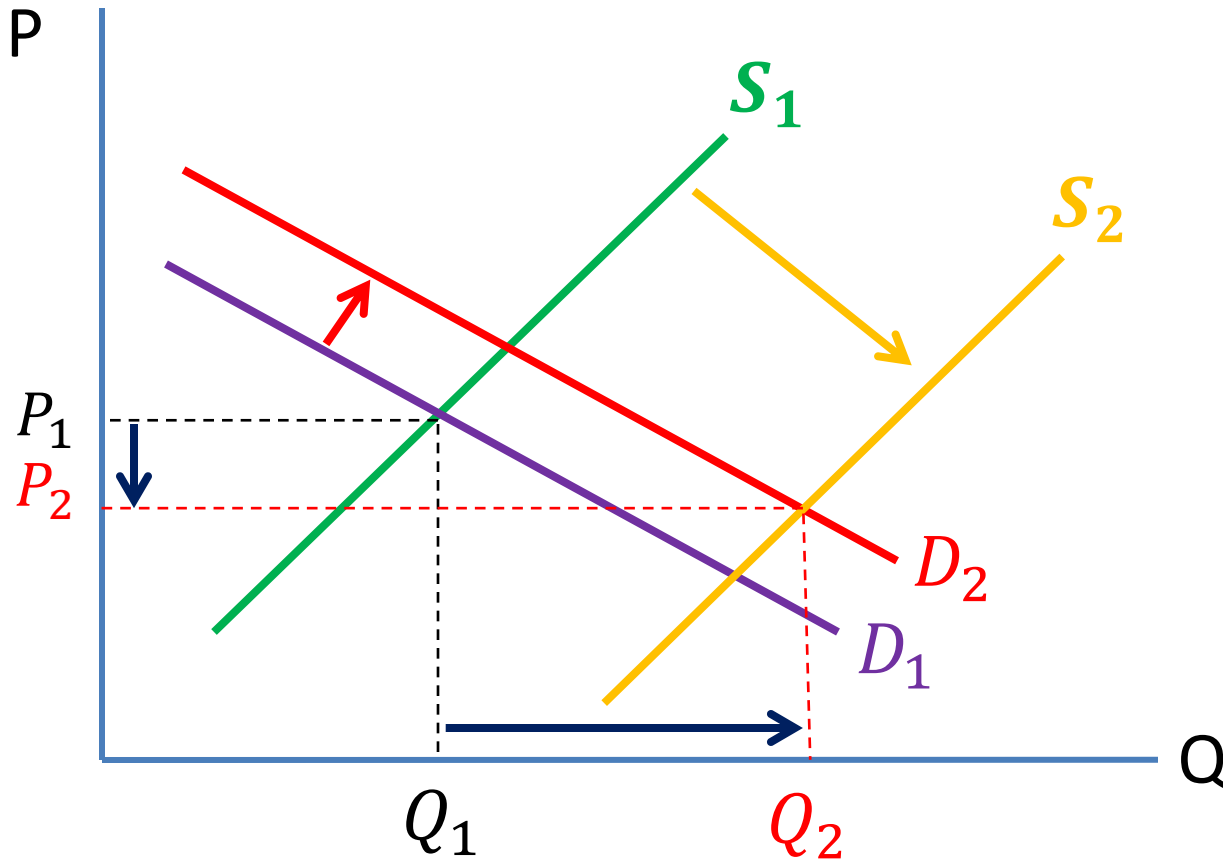
As a quick quiz, make sure you can explain at least a few of the entries in this table above using a supply-and-demand diagram.

## Example 1: An Increase in Demand and an Increase in Supply



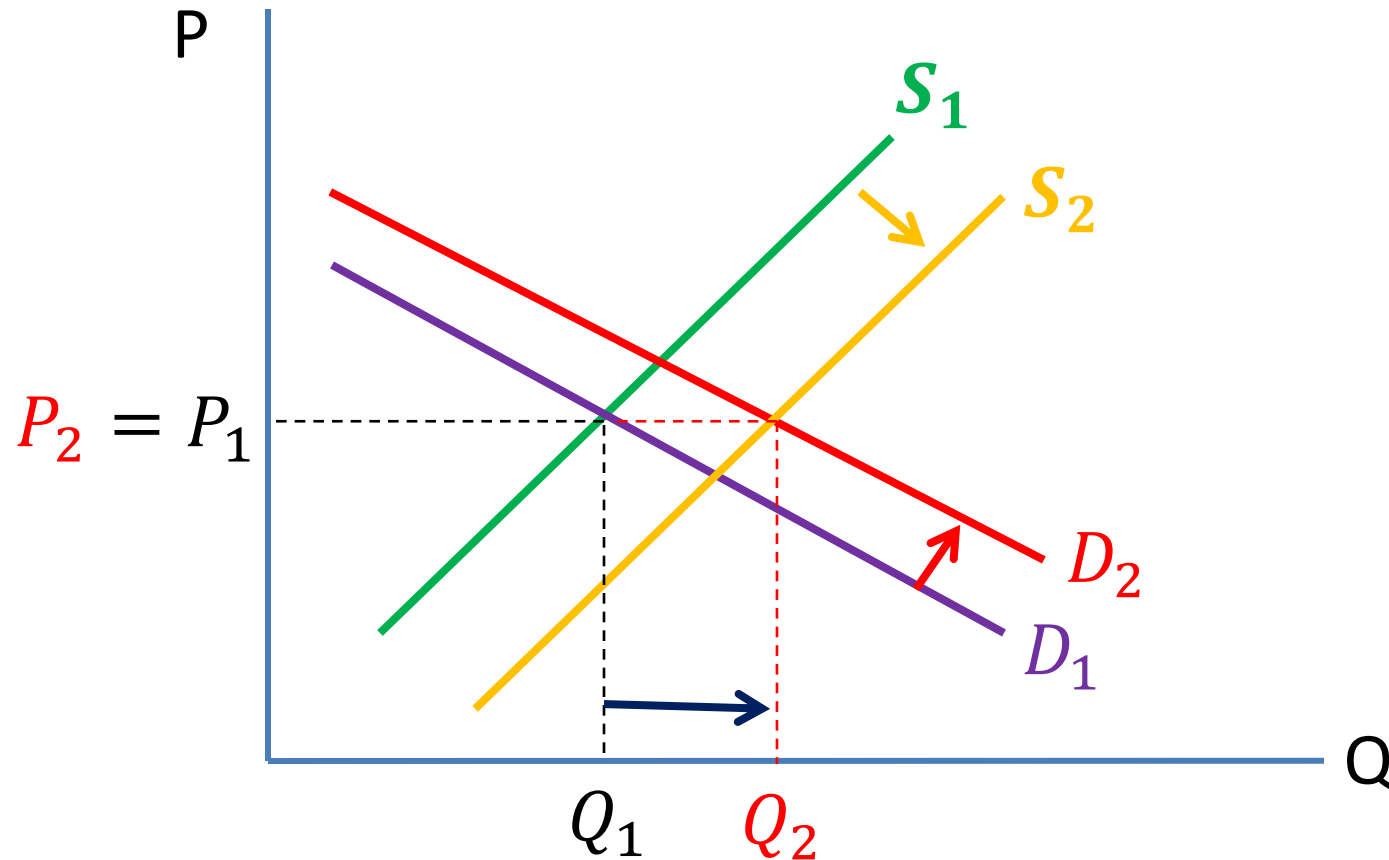
- a) **Case 1:** When the increase in demand is more than the increase in supply, then both  $Q$  and  $P$  go up.

## Example 1: An Increase in Demand and an Increase in Supply



**Case 2:** When the increase in supply is more than the increase in demand, then  $Q$  goes up and  $P$  goes down.

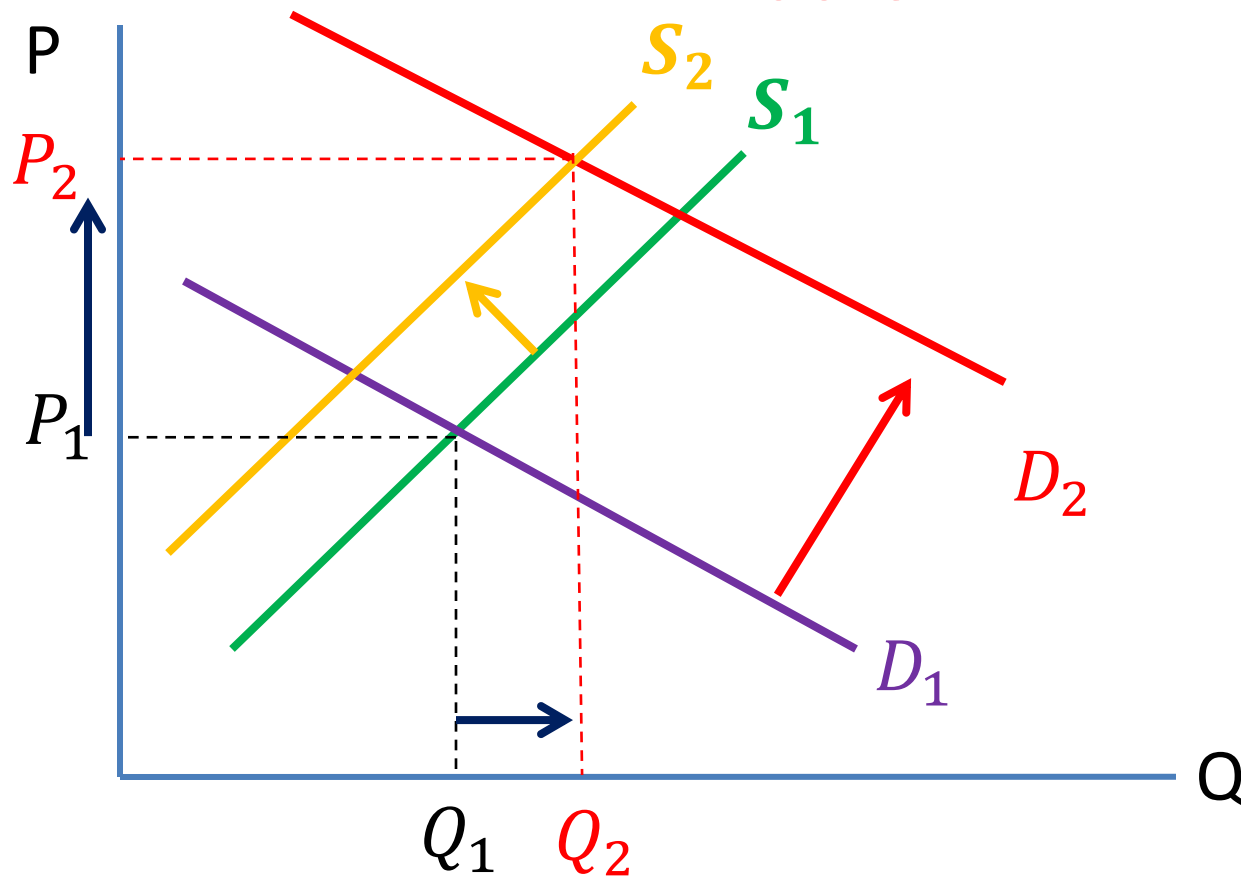
## Example 1: An Increase in Demand and an Increase in Supply



**Case 3:** When the increase in demand and supply is the same, then **P** remains the **same** and **Q** goes **up**.

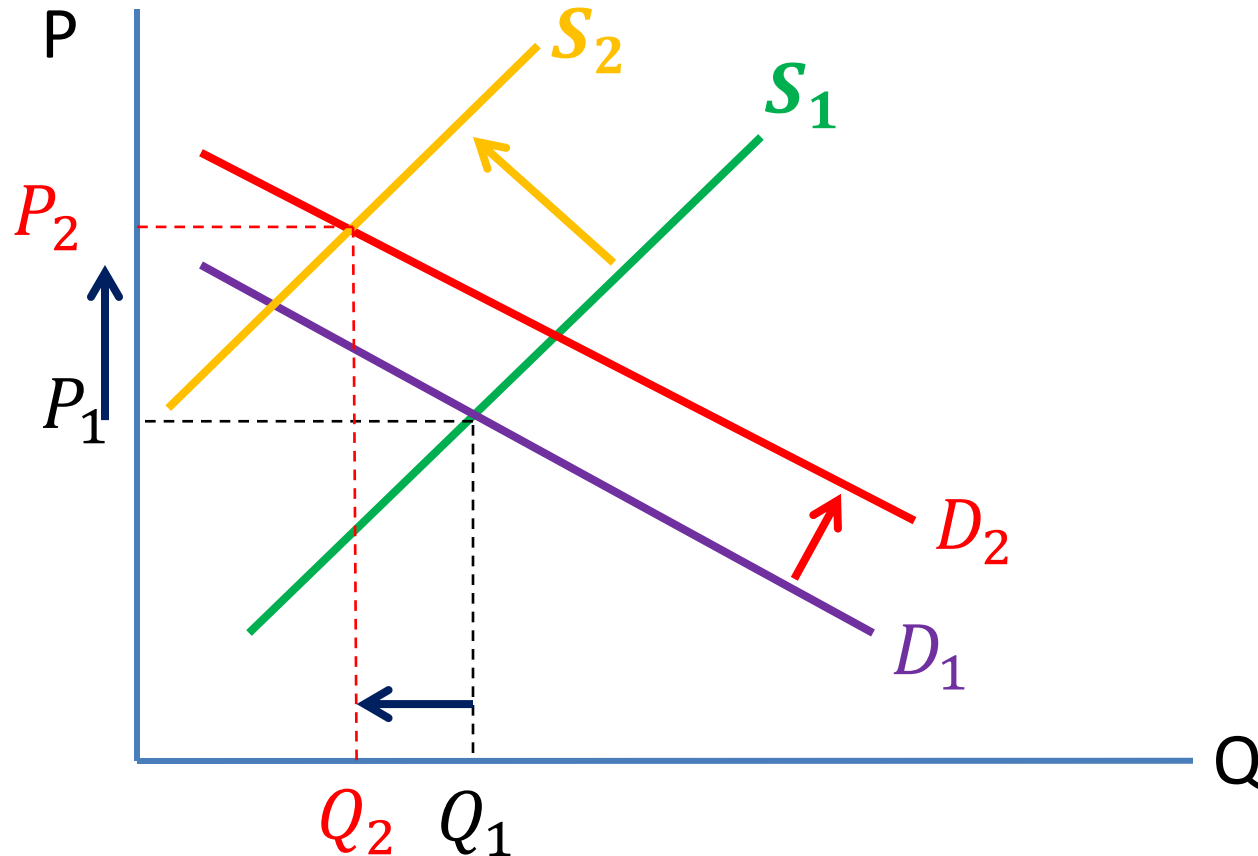


## Example 2: An Increase in Demand and a decrease in Supply



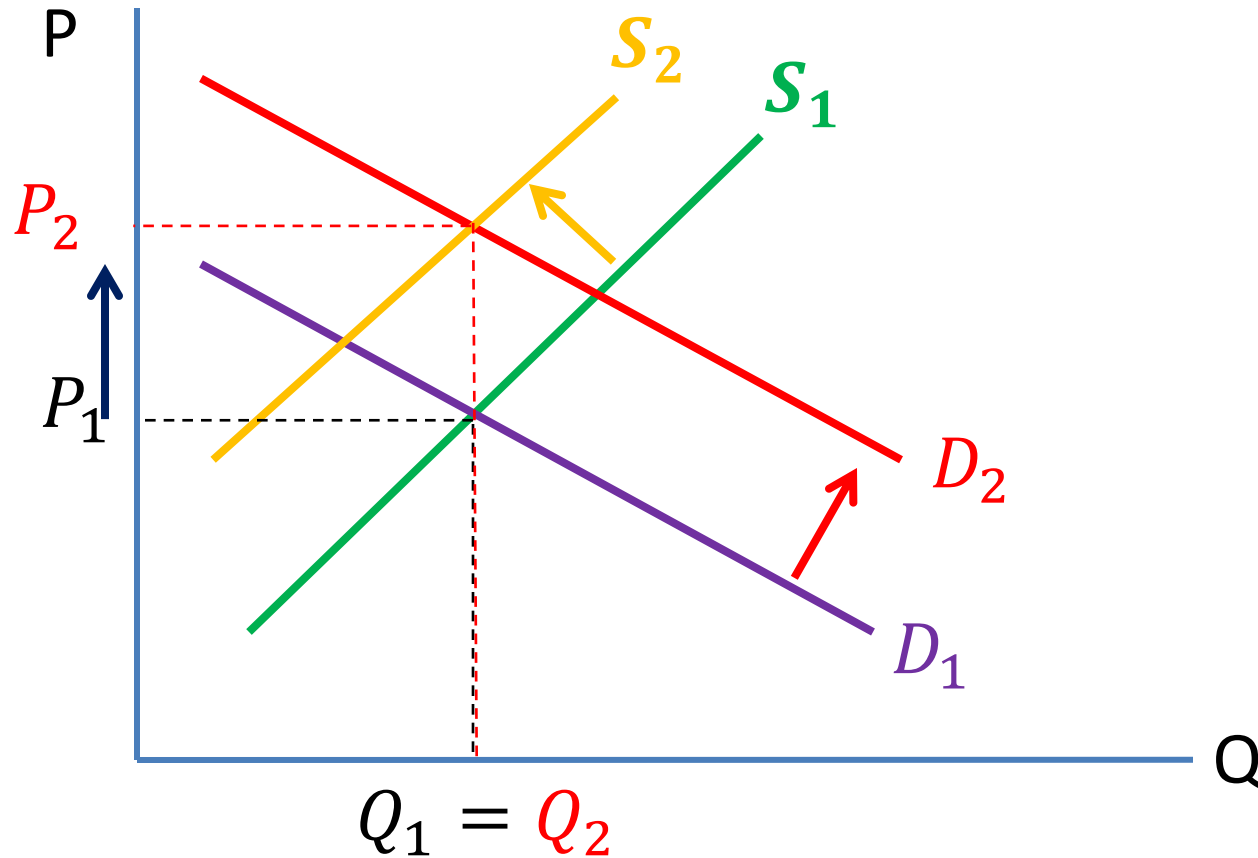
- a) **Case 1:** When the increase in demand is more than the decrease in supply, then both  $Q$  and  $P$  go up.

## Example 2: An Increase in Demand and a decrease in Supply



**b) Case 2:** When the increase in demand is less than the decrease in supply, then  $Q$  goes **down** and  $P$  goes **up**.

## Example 2: An Increase in Demand and a decrease in Supply



**c) Case 3:** When the increase in demand and the decrease in supply are equal, then  $Q$  remains **same** and  $P$  goes **up**.

## Exercises

1. The Effect of a **decrease** in **demand** and an **increase** in **supply** on equilibrium **Q** and **P**
  - a) **Case 1:** When the decrease in demand is more than the increase in supply
  - b) **Case 2:** When the decrease in demand is less than the increase in supply
  - c) **Case 3:** When the decrease in demand is equal to the increase in supply

## Exercises

2. The Effect of a **decrease** in **demand** and a **decrease** in **supply** on equilibrium **Q** and **P**
- a) **Case 1:** When the decrease in demand is more than the decrease in supply
  - b) **Case 2:** When the decrease in demand is less than the decrease in supply
  - c) **Case 3:** When the decrease in demand is equal to the decrease in supply

**End of Chap. 2**

**Good Luck!**